

Geometric Origin of 10^{11} in the Gravitational Constant

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Abstract

This note presents an empirical geometric observation: the number 10^{11} that appears in the gravitational constant G is not a decimal artifact. It arises naturally from four turns of a 3D golden spiral (similar to a Fibonacci spiral but rising in height) and the addition of a micro-step $1/720$.

1 Understanding 10^{11} in the Gravitational Constant

1.1 Basic spiral block

$$S = \phi^6 \times 31 \approx 556.272429$$

where $\phi = (1 + \sqrt{5})/2$ is the golden ratio.

1.2 Spiral inclination (logarithmic rise per turn)

$$\kappa = \log_{10}(S) \approx 2.745956$$

1.3 Four turns without micro-step

$$4\kappa \approx 10.983824$$

1.4 Target height ($\log_{10}(10^{11})$)

$$\log_{10}(10^{11}) = 11.000000$$

1.5 Difference

$$11.000000 - 10.983824 = 0.016176 \quad (\approx 0.147\% \text{ of } 11)$$

1.6 Micro-step

One full turn consists of 12 micro-steps, each contributing $1/720$:

$$\frac{12}{720} = \frac{1}{60} \approx 0.016667$$

This value almost exactly covers the gap of 0.016176.

1.7 Residual

$$0.016176 - 0.016667 = -0.000491 \quad (\approx 0.0045\% \text{ of } 11)$$

This small residual is interpreted as fractal noise of the system.

1.8 Resonant relation

$$\log_{10}(10^{11}) \approx 4 \times \log_{10}(\phi^6 \times 31) + \frac{12}{720}$$

Numerically: $10.983824 + 0.016667 = 10.999491$ (target 11.000000). The residual 0.000491 shows that 10^{11} is a resonant approximation, not an exact identity.

2 Equivalent forms of G_{Koch}

$$G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{\phi^{12} \times 10^{11}}$$
$$G_{\text{Koch}} = \frac{31^2 \sqrt{5}}{(144 + 1/720) \times 10^{11}}$$

Numerical value: $\approx 6.67354674496 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. Equivalence follows from the Kochenov Identity:

$$\frac{\phi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8})$$

3 Conclusion

The Universe is not a set of arbitrary constants. It is one spiral, read at different scales. This observation does not replace physics — it reveals that behind many measured constants lies a single spiral and a natural micro-step $1/720$.

4 Full precision reference

$$G_{\text{Koch}} = 6.6735467449616181237591997717904236079132174548715 \\ 1565557657672182332559409890102447913340675769273 \times 10^{-11}$$

Derived from: $G_{\text{Koch}} = 31^2 \sqrt{5} / (\phi^{12} \times 10^{11})$.

See also: *Kochenov Geometric Identity for G* (Zenodo, April 2026).